From singular tensor certificates to fixed testing features

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The strict-Gram construction preserves small tensor rank even when it changes the testing functions. We prove this fact, apply it to the all-H C1 certificate, and give explicit formulas for fixing the testing functions on a bounded source and testing region. The passage from these local regions to a full-set pencil remains to be constructed.

1 What the tensor criterion does and does not supply

Write $T = K_c \otimes_{\mathbb{R}} K_t$ for two labelled real closed extensions of $\mathbb{R}$, and let $B = B_c \otimes B_t$ be its bounded tensor subring. There is no disjointness assumption. For $z \in T$, $\operatorname{rank}_{\otimes}(z)$ is the smallest number of elementary source–testing products summing to z.

Scheiderer's Theorem 3.10 converts a universal bound d on the tensor rank of individual square roots into a bound d on the largest PSD block. It does not prescribe the number of blocks. Its proof constructs the algebra of testing functions at an individual pair, then makes two finite selections: a testing-image cover, combined by a disjoint union, and a support-admission cover, combined by a fibre product. See Lemmas 3.15–3.17, printed pp. 13–15 of arXiv:2004.04196v2. The dimensions and functions in those finite selections have not been evaluated in the present public Q222 source.

Thus there are three different finite objects to extract: a rank bound for each tensor root; a common finite testing space on one algebraic cover; and the multiplication tables that give the affine pencil. The first does not by itself print the other two.

2 A rank bound which survives the strict-Gram step

Lemma 1. *Let $G \in \operatorname{Sym}_r(B)$ have positive-definite real residue. Suppose $g_1, \dots, g_r \in K_c \otimes V$, where $V \subset K_t$ is an $\mathbb{R}$-linear space of dimension at most d. Then $g^{\mathsf{T}} G g$ is a finite sum of squares whose individual roots have tensor rank at most d. If only $\operatorname{rank}_{\otimes}(g_i) \le R$ is known, the bound rR follows.*

Proof. The scalar positive-residue lemma TF1 writes every positive-residue bounded tensor as a sum of squares of elementary tensors. This is stronger than merely giving arbitrary tensor square roots. The strict-Gram lemma TF2 first applies a real congruence and then expresses the quadratic form using positive-residue scalar coefficients times squares of real linear combinations of the g_i. Those combinations still lie in $K_c \otimes V$. Multiplication by an elementary tensor $a \otimes b$ places such a combination in $K_c \otimes bV$, whose real dimension is at most d. Expanding the scalar sums of squares therefore proves the bound. Taking V to be spanned by the testing factors in displayed rank-R expansions of all g_i gives $\dim V \le rR$. □

This proof needs no bound on the number of elementary tensors in a Gram coefficient. Such a bound affects the number of square roots, but not their individual tensor rank. The exact source is `03_tensor.tex`, TF1–TF2.

2.1 An evaluated rank bound for the all-H C1 theorem

Use the domain of the public theorem `thm:c1-unification-bounded-z-all-H`: $x = k^2$ and σ are positive source infinitesimals, $z > 0$ is bounded, h/σ is arbitrary, u, w are independent testing infinitesimals, and the original strict gaps and $D, \beta \geq 0$ are retained. No state ratio is imposed.

For its $\lambda > 0$, $WZ > 0$ branch, choose the common state sign so $W, Z > 0$. Every root in its main Gram belongs to the source-linear span of the following twelve testing functions:

$$\begin{aligned}\mathcal{H}_+ = \{&X, Y, Z, W, uZ, wZ, uW, uwW, u^2W,\\ &u^{3/2}W, u^{1/2}W, \sqrt{uwWZ}\}.\end{aligned} \tag{1}$$

These are distinct generalized monomials. Substituting squares for the positive variables makes them distinct ordinary monomials, so their real span has dimension exactly twelve on the open chart. Homogeneous state normalization does not change this independence.

Here is the entire root check. In the notation of that theorem,

root	testing factors needed
N_x	$X,\ uZ$
N_y	$Y,\ wZ$
R	$wZ,\ uW$
Γ_*	$Z,\ uwW$
A_θ	$W,\ u^2W$
T_θ	$W,\ uW$
$d_1, \dots, d_4$	$uZ,\ uwW,\ W,\ wZ,\ \sqrt{uwWZ}$
U_*, K_*	$u^{3/2}W,\ u^{1/2}W,\ uW,\ W.$

All omitted factors are pure source factors. In particular the four circuit monomials B_i are elementary tensors, and their geometric mean has testing factor $\sqrt{uwWZ}$. The twelve-root Gram is printed at lines 7550–7691. Lemma 1 therefore bounds its post-TF2 roots by twelve, rather than the coarser $12 \cdot 2$.

The retained squares do not enlarge this bound. Event and radial excess roots have rank at most three; the circuit remainder roots have rank at most two. For the angular budget, after expanding each $U + 1$ prefactor, the only root with more than two monomials is

$$\sqrt{a}\,U(V - Z_*) - (Z_* - UV)/\sqrt{a} = (\sqrt{a} + 1/\sqrt{a})UV - \sqrt{a}\,UZ_* - Z_*/\sqrt{a}.$$

It has rank at most three: a is pure source and U, V, Z_* are separated monomials. Multiplication by the other separated factors preserves that rank. Opposite-state-sign additions are separated nonnegative monomials. The zero-state-coordinate cases have the same or smaller ranks. In the $WZ \leq 0$ branch the $d_i = B_i$ do not even need the twelfth testing function.

For $\lambda \leq 0$, the Gram uses $(N_x, N_y, R, \Gamma, A_\theta, V)$ with $V = uw(u^2 - x)W$. Its testing span is contained in

$$\{X, Y, Z, W, uZ, wZ, uW, uwW, u^2W, u^3wW\},$$

of dimension ten. The source boundary $D = 0$ does not cause a rank loss problem: $\Gamma_* = \sqrt{4D/\sigma^2}\Gamma$ is then simply zero, and no inverse of D was used.

Proposition 2. *Conditional on the printed all-H C1 certificate, its entire stated local domain has a universal SOS representation with individual root tensor rank at most twelve, over all labelled real closed extensions, all allowed source faces and all states. This is a local tensor-rank statement. It is not a bound for the full quantum set, since the other source and projective-testing domains have not been included in this calculation.*

3 Where a fixed testing dictionary is still lost

The twelve functions in (1) span the roots before strict-Gram factorization. The Gram itself depends on the testing variables. It is therefore not a PSD matrix of source coefficients for those twelve functions. Treating it as one would omit exactly the state-normal and angular corrections being certified.

More explicitly, if $V = \operatorname{span}_{\mathbb{R}} \mathcal{H}_+$ and TF1 supplies the elementary root $a(c)b(t)$, a pre-Gram root in $K_c \otimes V$ is sent to $K_c \otimes bV$. This preserves the bound twelve on tensor rank, but does not in general preserve V. Different scalar coefficients may supply different bV. The proposition above asserts the rank bound only; it does not assert that all final roots lie in the twelve-function span.

The post-TF1 factors are also not automatically in one finite testing space. In TF4 the scalar splitting chooses a real $M > |y|$ after the bounded testing element y is given. Its testing square roots include $\sqrt{M+y}$ and $\sqrt{M-y}$. The real congruence and the positive scalar budgets may likewise be chosen separately at each source/test pair.

Lemma 3. *For distinct real $M_j > 1$, the functions $\sqrt{M_j + t}$ on $-1 < t < 1$ are linearly independent over $\mathbb{R}$. Consequently the union of the allowable TF1 choices is not a finite-dimensional testing space, even for a linear testing factor.*

Proof. In $\mathbb{R}(t)^*/\mathbb{R}(t)^{*2}$ the classes of $M_j + t$ are independent: a product of a nonempty subset has odd valuation at each of its distinct zeros $t = -M_j$. The associated multiquadratic function field therefore has an automorphism changing the sign of one chosen square root and fixing the others. Apply that automorphism to a supposed real linear relation and subtract it from the original relation. Its chosen coefficient is zero. Repeat for each index. A relation between the analytic positive branches on an interval would also be a function-field relation, so the same conclusion applies to the displayed functions. □

This is an obstruction to taking the union of the displayed pointwise roots, not an obstruction to any possible finite lift. A different certificate can fix M on a uniformly bounded box. Indeed, on a fixed rectangle $|x(c)| \le \varepsilon/M$, $|y(t)| \le M$, with fixed real $\varepsilon, M > 0$, the identity

$$\varepsilon + x(c)y(t) = \tfrac{1}{2}\big[(\varepsilon/M + x(c))(M + y(t)) + (\varepsilon/M - x(c))(M - y(t))\big]$$

has nonnegative separated factors. For this channel the only testing radicals are the two fixed functions $\sqrt{M + y(t)}$ and $\sqrt{M - y(t)}$. Equality on either boundary is harmless. Thus multiplication of a fixed d-function testing span uses at most $2d$ functions for this channel. The unresolved step is to supply finitely many such rectangles and prove their coverage, not the scalar identity on any one of them. An auxiliary algebraic cover can also change the available functions, but then one must prove a global SOS identity on that fixed cover; an existential choice of its new coordinates at each source/test pair is insufficient.

There is a second issue at precisely the same interface. The manuscript's conditions "infinitesimal", "bounded unit" and "unbounded" refer to types over $\mathbb{R}$; they are not fixed real semialgebraic boxes. A finite list of such source cases is not yet an evaluated finite list of real source/test rectangles with fixed constants. Qualitative real-spectrum compactness can select those data if its universal premise is valid. A compiler must actually supply them.

4 A fixed-box compilation which removes that obstruction

The following statement gives an explicit target for completing one local box, rather than adjoining all possible TF1 radicals.

Lemma 4 (Fixed testing radicals)**.** *Let $h_1, \dots, h_d$ be fixed testing functions on Y. Suppose a finite source-dependent quadratic form has been written as a sum of scalar channels times squares of source-linear combinations of the h_i. In every channel suppose*

$$q(c,t) = q_0(c) + \sum_{j=1}^{m} a_j(c) b_j(t), \qquad q_0(c) \geq c_0 > 0, \quad |a_j(c)| \leq A,$$

where $m \geq 1$, the b_j are a fixed list of testing functions and $c_0, A > 0$ are fixed rational numbers. Put $\gamma = c_0/(mA)$ and assume $|b_j(t)| \leq \gamma$ on Y. Then a common testing spanning list is

$$\{h_i\}_{i=1}^{d} \ \cup \ \{h_i\sqrt{\gamma + b_j},\ h_i\sqrt{\gamma - b_j} : 1 \leq i \leq d, 1 \leq j \leq m\},$$

with at most $d(1+2m)$ functions. All scalar multipliers are nonnegative source factors. The formula is an identity over all labelled real closed extensions satisfying the same finite bounds.

Proof. Exactly

$$q = (q_0 - c_0) + \frac{1}{2}\sum_{j=1}^{m}\big[(A + a_j)(\gamma + b_j) + (A - a_j)(\gamma - b_j)\big].$$

All four factors in every bracket are nonnegative in their own fields. Take their separate square roots and multiply by the channel's source-linear combination of h_i. The source-only term uses no new testing function. The proposed list follows. □

For the bounded-z C1 family, the accompanying *Fixed testing functions at singular C1 contacts* carries out this construction on the independent rectangles

$$0 < x, \sigma \leq 2^{-160}, \qquad 0 < z \leq 1, \qquad |u|, |w| \leq 2^{-100},$$

with the original source gaps and $D, \beta \geq 0$. The ratio $H = h/\sigma$ is unrestricted. In the positive-angular case, twelve main roots and seven nonconstant testing monomials give $12(1 + 2 \cdot 7) = 180$ functions by Lemma 4. The retained event, angular and circuit squares enlarge this list to 220. A source-only shear treats the nonpositive-angular case with 157 functions. Thus their union contains at most 377 functions for the complete real-state polynomial on this rectangle.

The two constructions print every source bound, coefficient matrix, square factor and testing function. They include both testing axes, all real state signs and the permitted zero event weights. The JSON files supply the labelled lists and coefficient prescriptions. Extending these constructions across the remaining source and testing domains is the outstanding global selection problem.

5 The final finite matrix once a common cover is evaluated

Suppose one has actually printed a finitely generated real algebra A, a visible map $\phi : Y(\mathbb{R}) \to \mathbb{R}^8$, and finite subspaces $U_1, \dots, U_s \subset A$ such that

$$\mathcal{Q}_{222} \subset \phi(Y(\mathbb{R})), \qquad \phi^*(P) \subset \mathbb{R}_{\geq 0} + \sum_{i=1}^{s} \Sigma U_i^2,$$

where P is the full cone of nonnegative affine quantum slacks. Both inclusions must be proved for those actual data. Take bases $v_{i1}, \dots, v_{id_i}$ and a basis of

$$W = \operatorname{span}\{1, \phi_1, \dots, \phi_8, v_{i\alpha} v_{i\beta} : i, \alpha, \beta\}.$$

For $\ell \in W^*$ impose

$$\ell(1) = 1, \quad \ell(\phi_\nu) = q_\nu, \qquad M_i(\ell) = [\ell(v_{i\alpha} v_{i\beta})]_{\alpha,\beta} \succeq 0 \quad (1 \le i \le s).$$

The printed product coefficients in the basis of W are exactly the affine-pencil entries. Evaluation at a covering point proves the forward inclusion. If a feasible q were outside the compact convex quantum set, a nonnegative affine slack f with $f(q) < 0$ would give $\ell(\phi^* f) \ge 0$ from the SOS identity, a contradiction. This proves the reverse inclusion without replacing the required global SOS identity by tests at finitely many real sources.

This construction has total block order $N = \sum_i d_i$ and at most $\dim W - 9$ auxiliary coordinates (the full quantum set has affine dimension eight). Its entries become evaluated only when the cover, features and product basis are evaluated. Neither the local rank bound twelve nor the 377-function local union supplies the missing global list $U_1, \dots, U_s$.